

§7.7 - Approximate Integration

Problem

We want to approximate

$$\int_a^b f(x) \, dx.$$

Definition #1 (Uniform Partition)

Let $[a, b] \subset \mathbf{R}$, $n \in \mathbf{N}$ and

$$\Delta x = \frac{b - a}{n}.$$

Then

$$P = \{x_0, x_1, \dots, x_n\}$$

where

$$x_k = a + k\Delta x = x_{k-1} + \Delta x$$

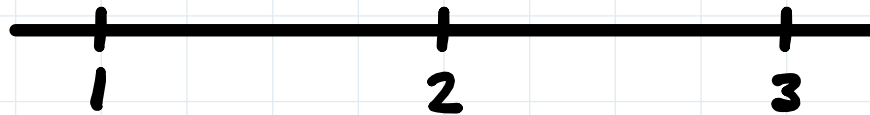
is a **uniform partition** of $[a, b]$ with **step-size** Δx .

If $a = 1$ and $b = 3$, then

$$\Delta x = \frac{3-1}{n} = \frac{2}{n}$$

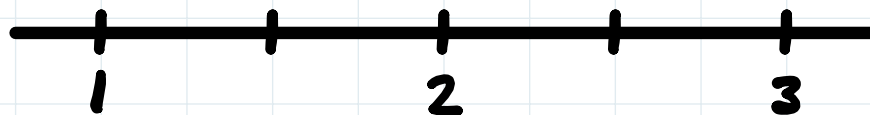
For $n = 2$, $\Delta x = 1$

$$P = \{1, 2, 3\}$$



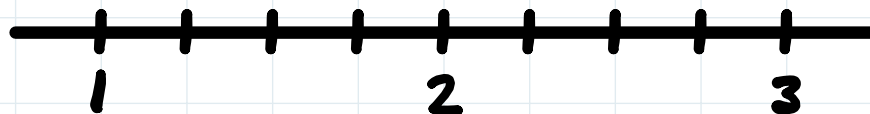
For $n = 4$, $\Delta x = 1/2$

$$P = \{1, 3/2, 2, 5/2, 3\}$$



For $n = 8$, $\Delta x = 1/4$

$$P = \{1, 5/4, \dots, 11/4, 3\}$$



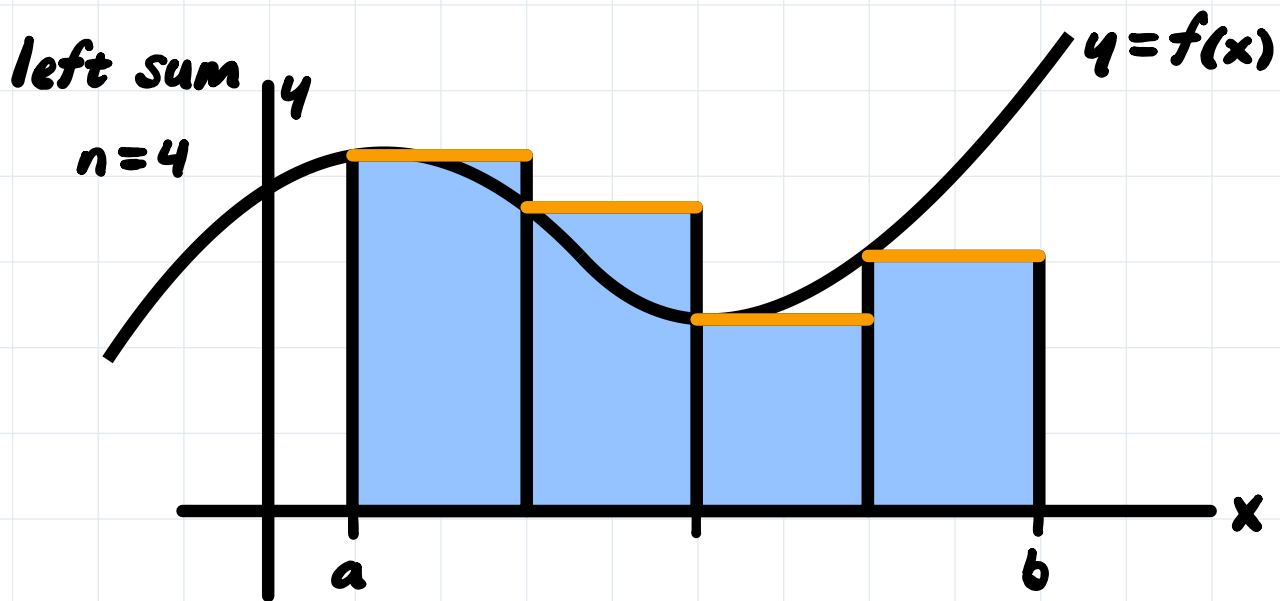
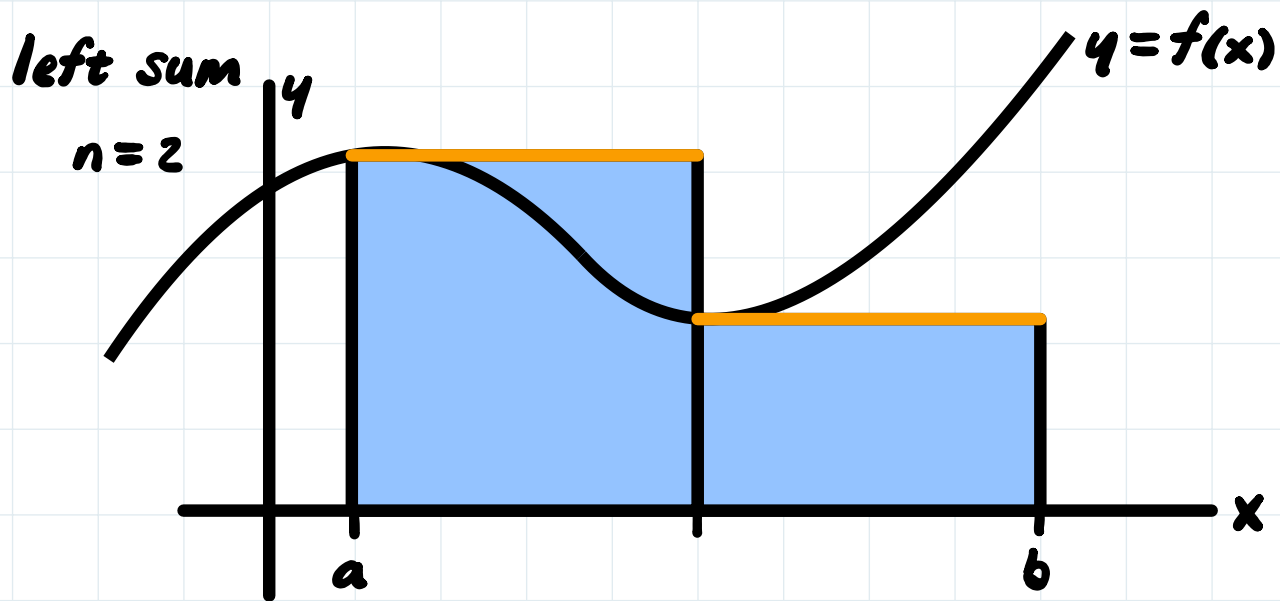
Definition #2 (Left Riemann Sum)

Assume $[a, b]$ is an interval, $\Delta x = (b - a)/n$, and f is integrable on $[a, b]$.

Then

$$\begin{aligned} L_n &= \sum_{k=1}^n f(x_{k-1}) \Delta x \\ &= (f(x_0) + f(x_1) + \dots + f(x_{n-1})) \Delta x \end{aligned}$$

is a **left Riemann sum** approximation of $\int_a^b f$.



With $n=4$, $\Delta x = 1/2$

$$\int_1^3 f(x) dx \approx (f(1) + f(3/2) + f(2) + f(5/2)) \frac{1}{2}.$$

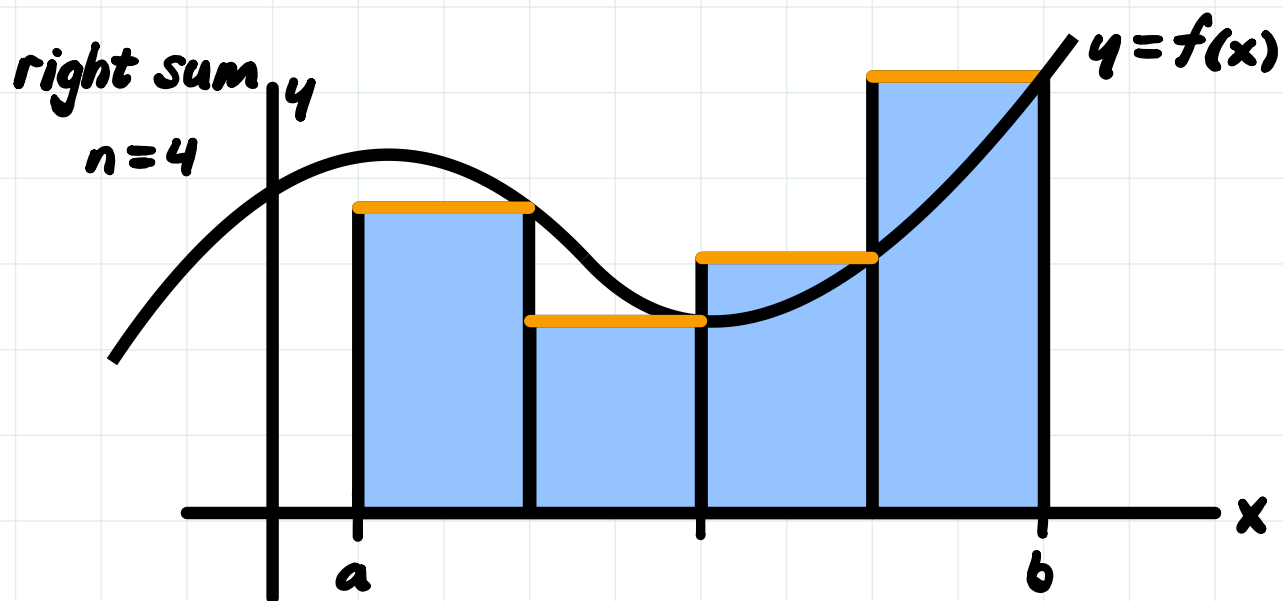
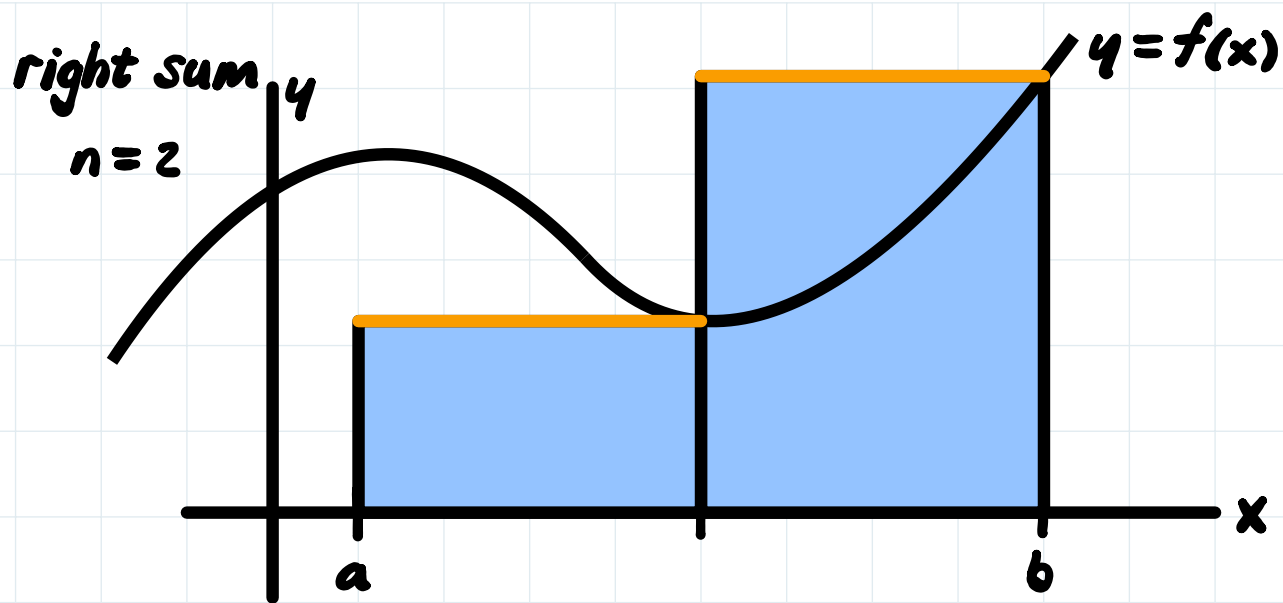
Definition #3 (Right Riemann Sum)

Assume $[a, b]$ is an interval, $\Delta x = (b - a)/n$, and f is integrable on $[a, b]$.

Then

$$\begin{aligned} R_n &= \sum_{k=1}^n f(x_k) \Delta x \\ &= (f(x_1) + \dots + f(x_n)) \Delta x \end{aligned}$$

is a **right Riemann sum** approximation of $\int_a^b f$.



With $n=4$, $\Delta x = 1/2$

$$\int_1^3 f(x) dx \approx (f(3/2) + f(2) + f(5/2) + f(3)) \frac{1}{2}.$$

Remark

We have

$$L_n - R_n = (f(x_0) - f(x_n))\Delta x = (f(a) - f(b))\Delta x$$

and

$$\int_a^b f(x) \, dx \approx L_n \approx R_n.$$

Definition #4 (Midpoint Riemann Sum)

Assume $[a, b]$ is an interval, $\Delta x = (b - a)/n$, and f is integrable on $[a, b]$.

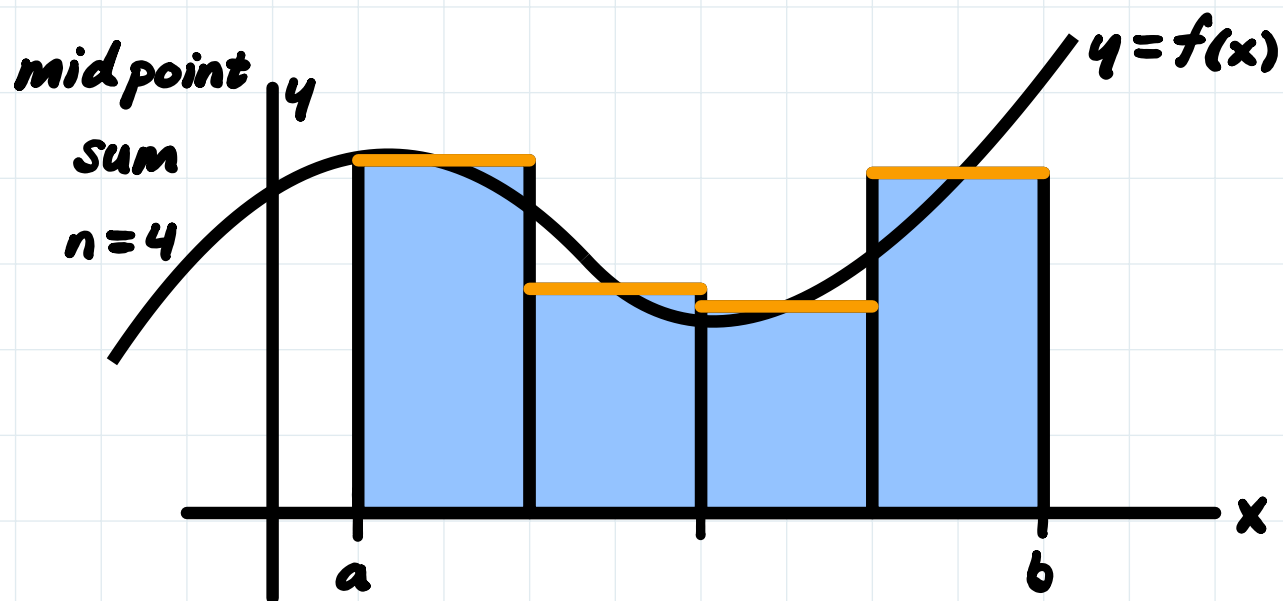
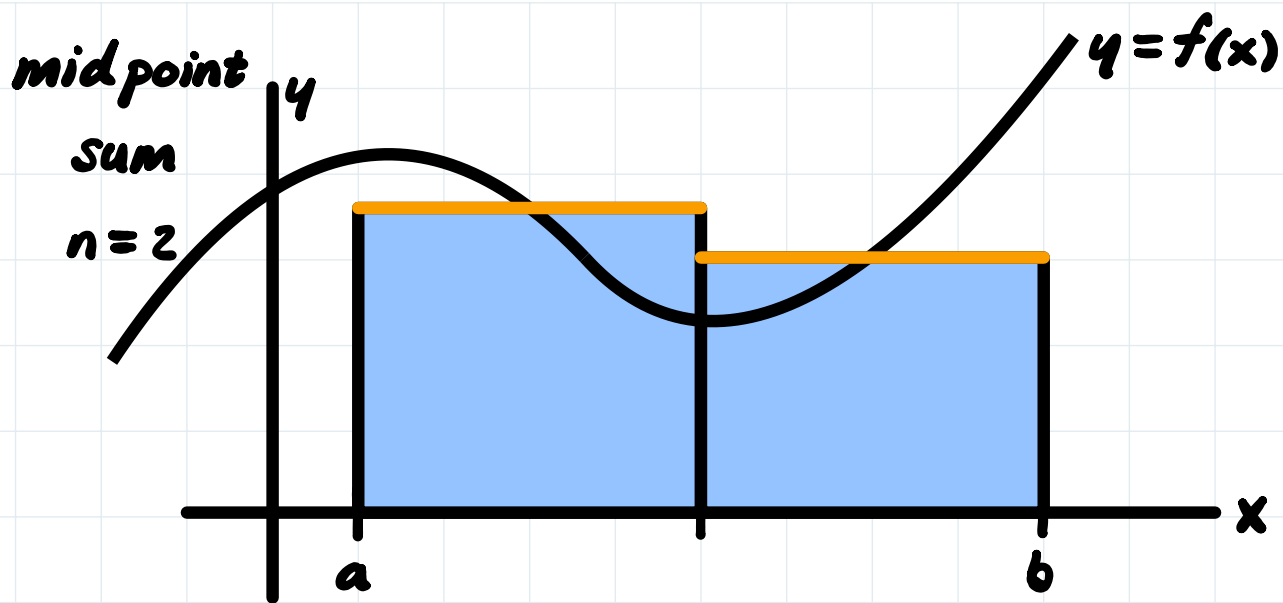
Then

$$M_n = \sum_{k=1}^n f(\bar{x}_k) \Delta x = (f(\bar{x}_1) + \dots + f(\bar{x}_n)) \Delta x$$

where

$$\bar{x}_k = x_{k-1} + \Delta x/2 = \frac{x_{k-1} + x_k}{2}$$

is a **midpoint Riemann sum** approximation of $\int_a^b f$.



With $n = 4$, $\Delta x = 1/2$

$$\int_1^3 f(x) dx \approx (f(5/4) + f(7/4) + f(9/4) + f(11/4)) \frac{1}{2}.$$

We are **not** reusing the positions found in the left/right sums.

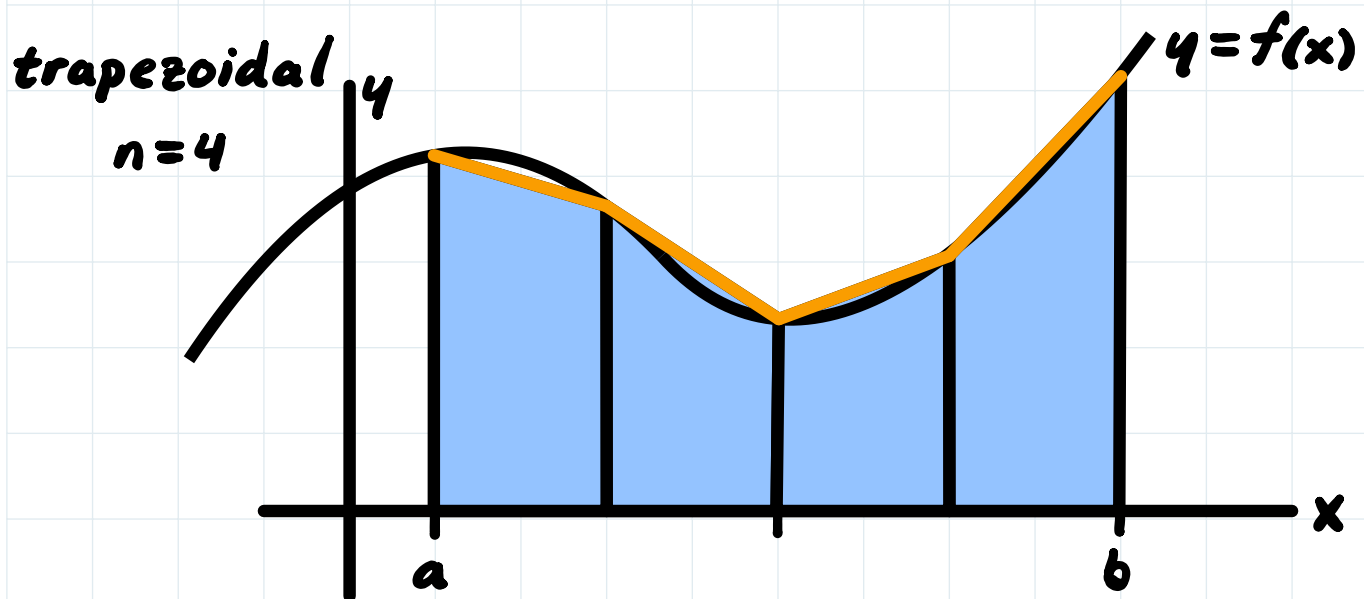
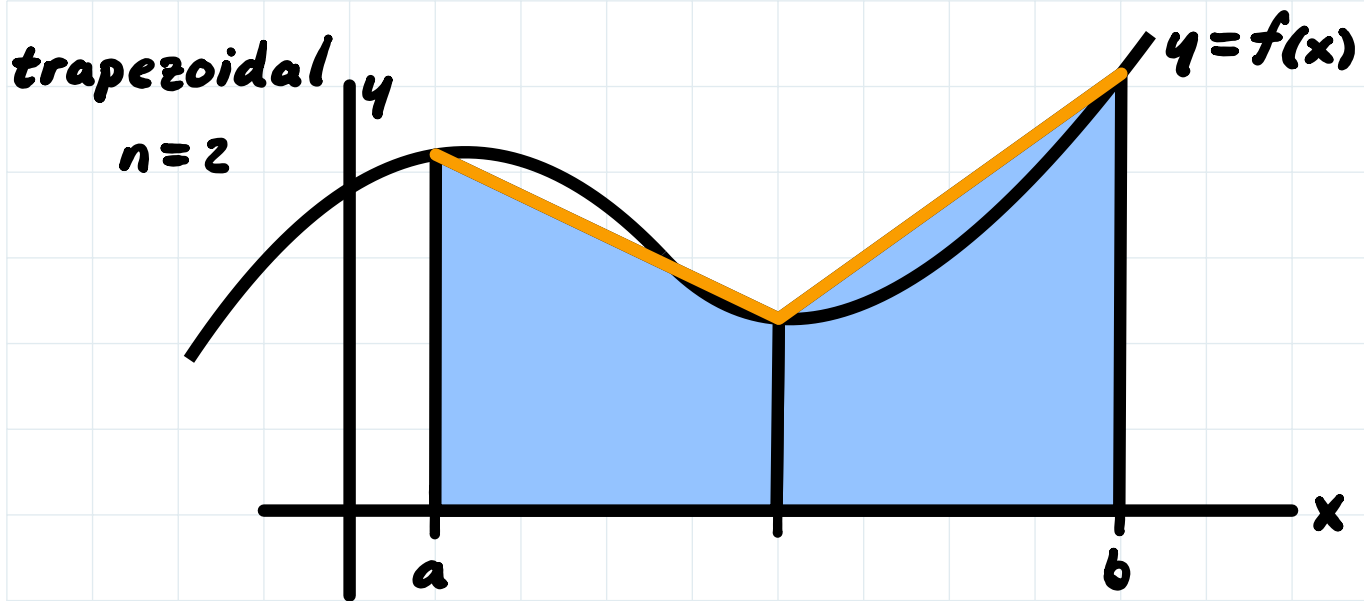
Definition #5 (Trapezoidal Rule)

Assume $[a, b]$ is an interval, $\Delta x = (b - a)/n$, and f is integrable on $[a, b]$.

Then

$$\begin{aligned} T_n &= \left(\frac{f(x_0)}{2} + \sum_{k=1}^{n-1} f(x_k) + \frac{f(x_n)}{2} \right) \Delta x \\ &= (f(x_0) + 2f(x_1) + 2f(x_2) + \dots + 2f(x_{n-1}) + f(x_n)) \frac{\Delta x}{2} \end{aligned}$$

is a **trapezoidal rule** approximation of $\int_a^b f$.



With $n=4$, $\Delta x = 1/2$

$$\int_1^3 f(x) dx$$

$$\approx (f(1) + 2f(3/2) + 2f(2) + 2f(5/2) + f(3)) \frac{1}{4}.$$

We **are** reusing the positions found in the left/right sums.

Remark

Naturally,

$$\int_a^b f(x) \, dx \approx T_n$$

and

$$T_n = \frac{L_n + R_n}{2},$$

but T_n is **not** a Riemann sum.

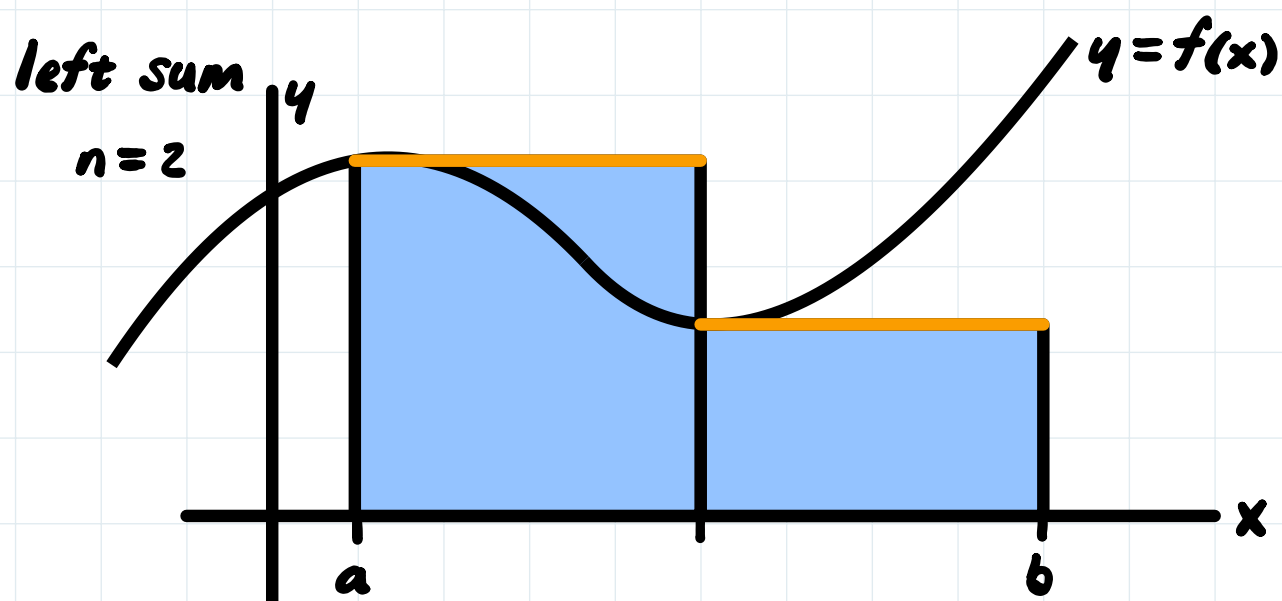
We are not estimating the area by rectangles, so the Trapezoidal Rule is an approximation of a different kind!

Left/right Riemann sums will be exact if f is constant over our partition.

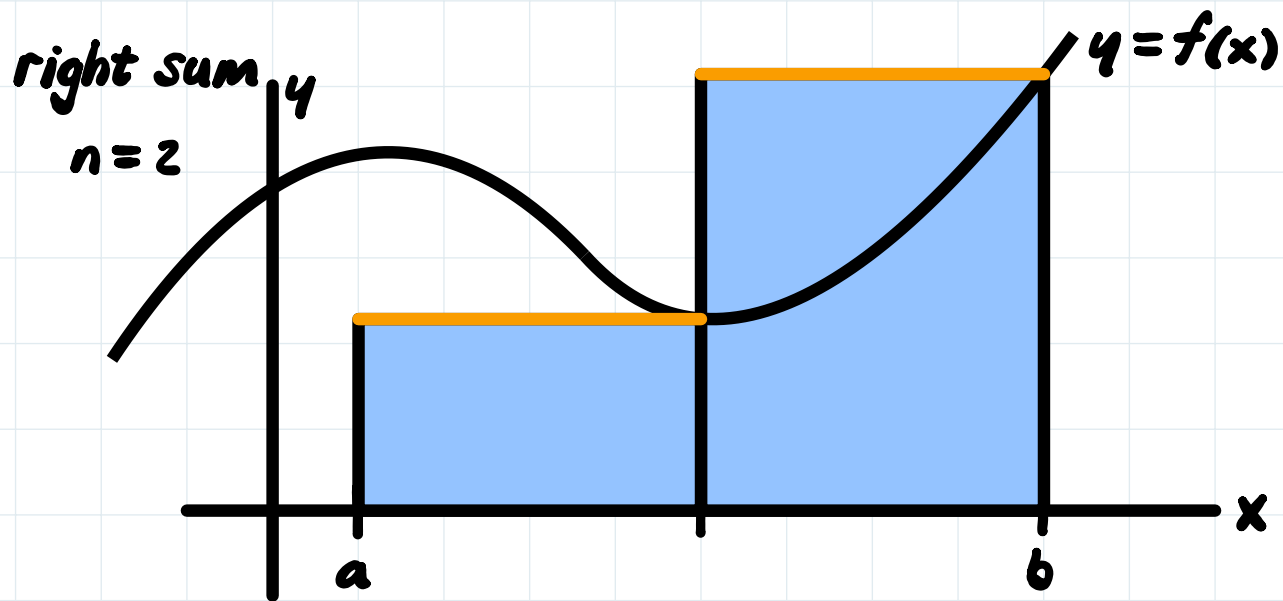
Trapezoidal Rule will be exact if f is piecewise linear over our partition.

If $f(x) \geq 0$ on $[a, b]$:

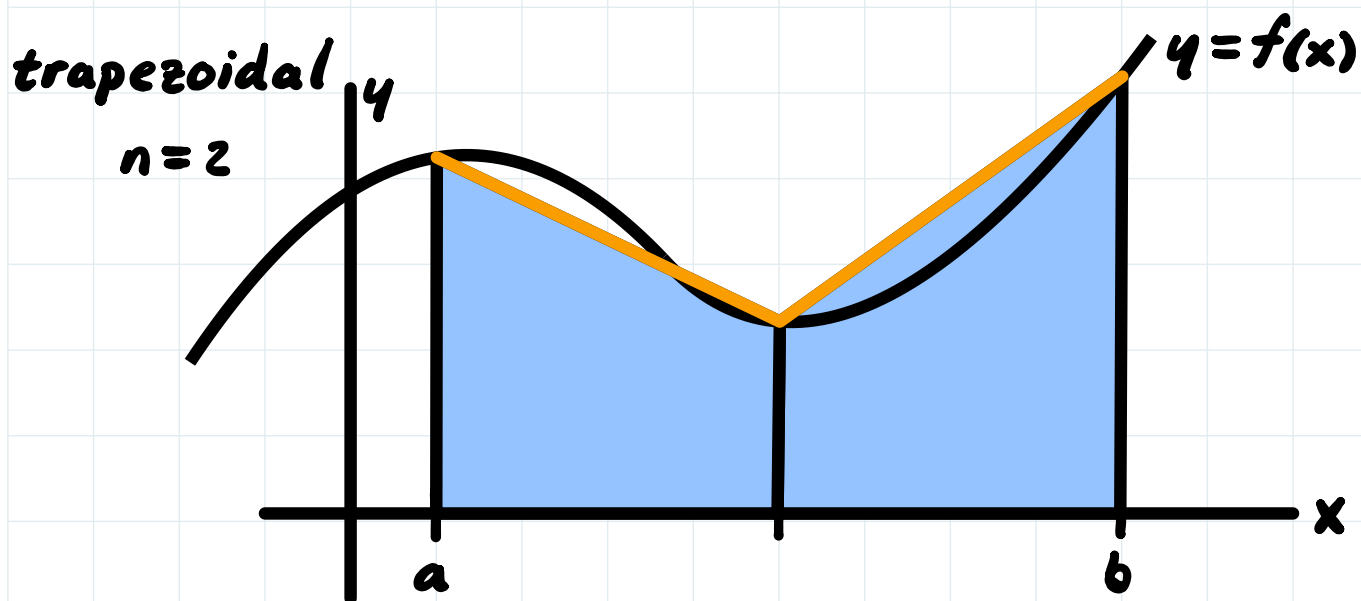
left sum: over estimate if $f' < 0$, f decreasing



right sum: over estimate if $f' > 0$, f increasing



trapezoidal: over estimate if $f'' > 0$, f concave up



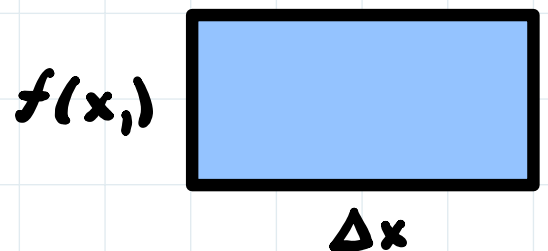
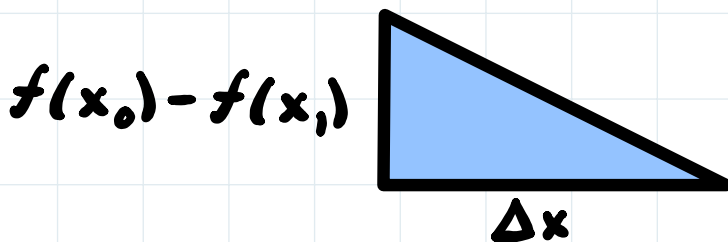
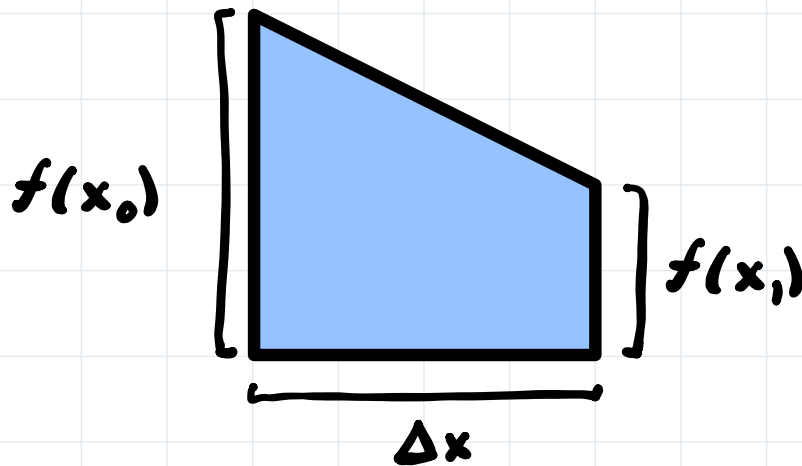
Simple case $n = 1: \Delta x = b - a$

left sum: $f(x_0)\Delta x$

right sum: $f(x_1)\Delta x$

midpoint sum: $f\left(x_0 + \frac{\Delta x}{2}\right)\Delta x$

trapezoidal sum: $\frac{1}{2}(f(x_0) + f(x_1))\Delta x$



Total area of the trapezoid:

$$f(x_1)\Delta x + \frac{1}{2}(f(x_0) - f(x_1))\Delta x = \frac{1}{2}(f(x_0) + f(x_1))\Delta x$$

Example #1

Compute

$$T_2, \quad M_2, \quad \text{and} \quad T_4$$

for

$$\int_0^2 e^x \, dx.$$

We have

$$a = 0, \quad b = 2$$

and for $n = 2$, $\Delta x = 1$.

Then

$$T_2 = (e^0 + 2e^1 + e^2) \frac{1}{2} \approx 6.91281$$

$$M_2 = (e^{1/2} + e^{3/2}) \cdot 1 \approx 6.13041$$

while

$$T_4 = (e^0 + 2e^{1/2} + 2e^1 + 2e^{3/2} + e^2) \frac{1}{4} \approx 6.52161$$

Note:

$$\int_0^2 e^x dx \approx 6.38906$$

Definition #6 (Error Terms)

Assume M_n and T_n are the midpoint and trapezoidal rule approximations of $\int_a^b f$.

Then

$$E_M(n) = \int_a^b f(x) \, dx - M_n \quad \text{and} \quad E_T(n) = \int_a^b f(x) \, dx - T_n$$

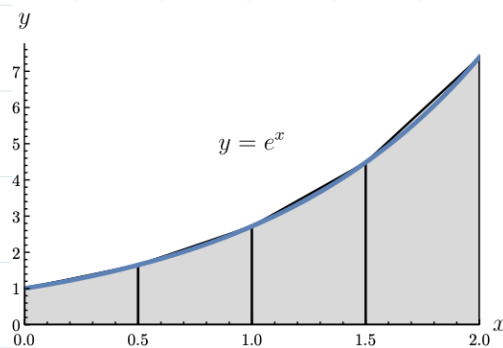
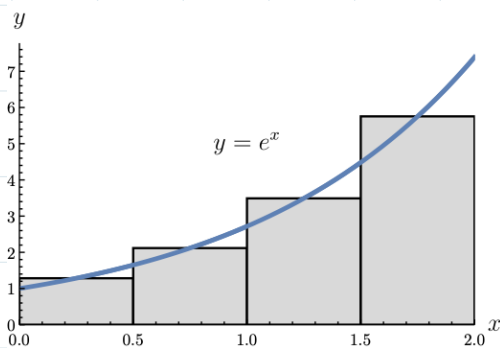
are the **error terms** of the approximations.

We have

$$\begin{aligned} E_T(2) &\approx 6.38906 - 6.91281 \\ &= -0.52375 \end{aligned}$$

$$\begin{aligned} E_M(2) &\approx 6.38906 - 6.13041 \\ &= 0.25865 \end{aligned}$$

An overestimate for Trapezoid, and an underestimate for Midpoint.



Theorem #1

Let f'' be continuous on $[a, b]$ and assume M_n and T_n are midpoint and trapezoidal rule approximations of $\int_a^b f$.

Then

$$|E_M(n)| \leq \frac{K(b-a)^3}{24n^2} = \frac{K(b-a)}{24}(\Delta x)^2$$

and

$$|E_T(n)| \leq \frac{K(b-a)^3}{12n^2} = \frac{K(b-a)}{12}(\Delta x)^2$$

where $|f''(x)| \leq K$ for all $x \in [a, b]$.

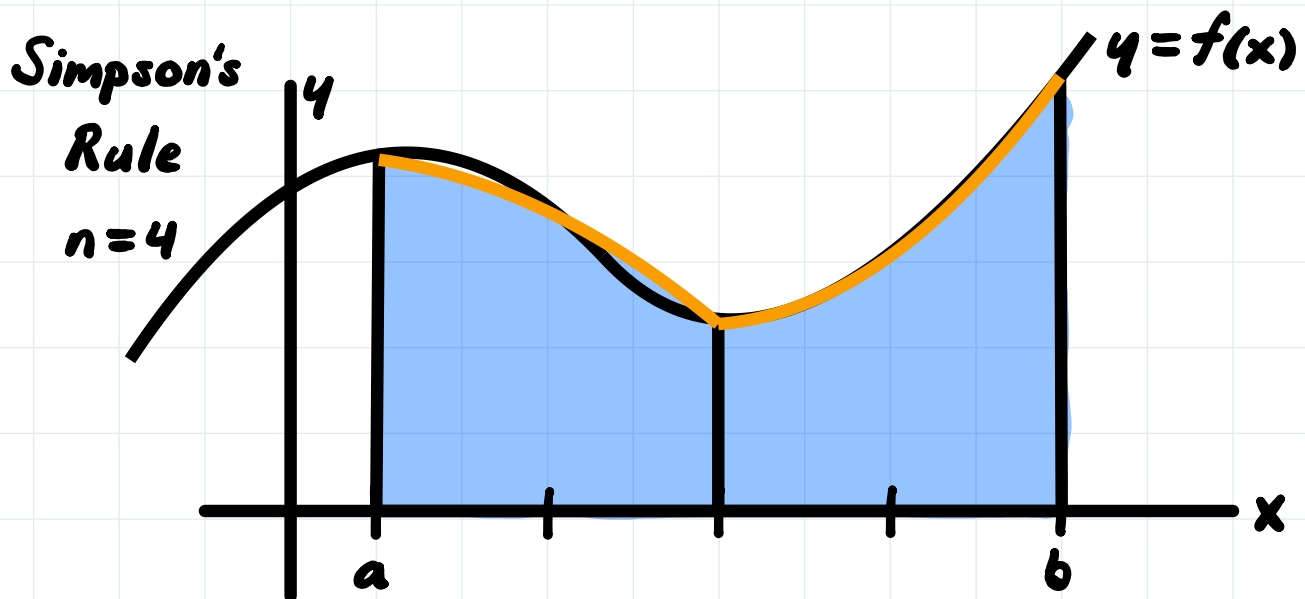
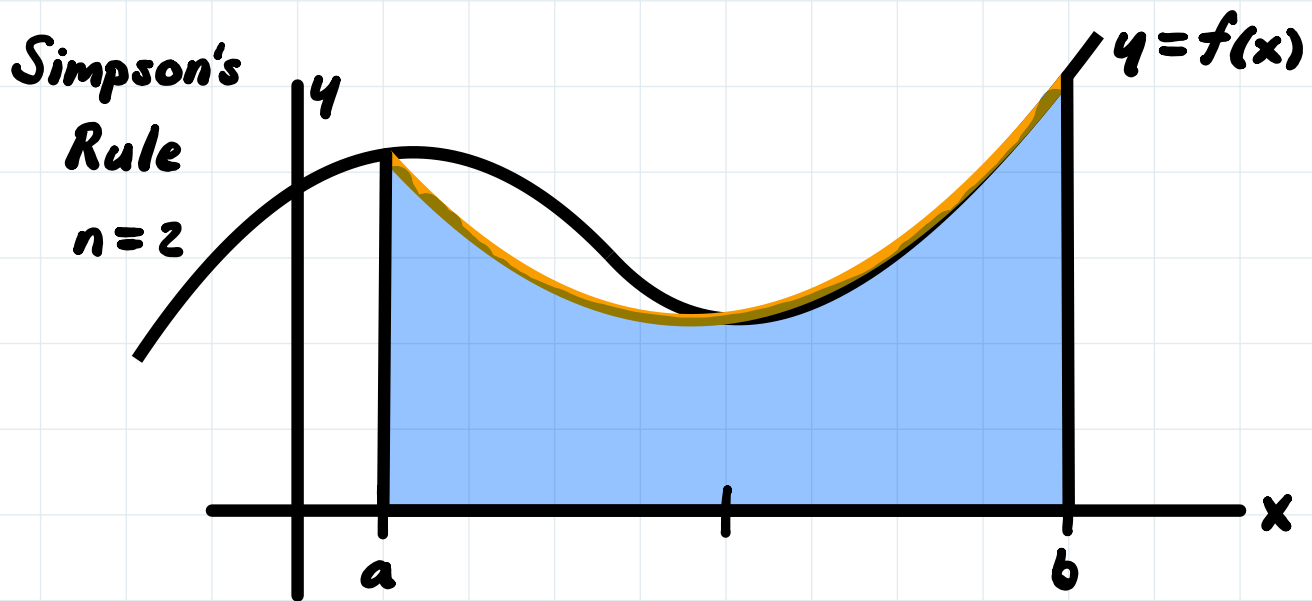
Definition #7 (Simpson's Rule)

Assume $[a, b]$ is an interval, n is even, $\Delta x = (b - a)/n$, and f is integrable on $[a, b]$.

Then

$$\begin{aligned} S_n = & (f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + \dots \\ & + 4f(x_{n-3}) + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)) \frac{\Delta x}{3} \end{aligned}$$

is a **Simpson's rule** approximation of $\int_a^b f$.



Unlike the trapezoidal rule, Simpson's rule approximates f by a quadratic polynomial.

Simple case: $n = 2$ $\Delta x = \frac{b-a}{2}$

left: $(f(x_0) + f(x_1))\Delta x$

right: $(f(x_1) + f(x_2))\Delta x$

midpoint:

$$\left(f\left(x_0 + \frac{\Delta x}{2}\right) + f\left(x_1 + \frac{\Delta x}{2}\right) \right) \Delta x$$

trapezoidal:

$$\begin{aligned} T_2 &= (f(x_0) + f(x_1)) \frac{\Delta x}{2} + (f(x_1) + f(x_2)) \frac{\Delta x}{2} \\ &= (f(x_0) + 2f(x_1) + f(x_2)) \frac{\Delta x}{2} \end{aligned}$$

Simpson's

$$S_2 = (f(x_0) + 4f(x_1) + f(x_2)) \frac{\Delta x}{3}$$

while

$$\begin{aligned} S_4 &= (f(x_0) + 4f(x_1) + f(x_2)) \frac{\Delta x}{3} \\ &\quad + (f(x_2) + 4f(x_3) + f(x_4)) \frac{\Delta x}{3} \\ &= (f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)) \frac{\Delta x}{3}. \end{aligned}$$

Consider Simpson's Rule over the partition

$$P = \{-\Delta x, 0, \Delta x\}.$$

Let

$$f(x) \approx p(x) = A_2 x^2 + A_1 x + A_0$$

with

$$f(x_0) = p(x_0), \quad f(x_1) = p(x_1), \quad f(x_2) = p(x_2).$$

We expect

$$\int_{x_0}^{x_2} p(x) dx \approx \int_{x_0}^{x_2} f(x) dx$$

since f and p are equal at the partition points.

We have

$$\begin{aligned}
\int_{x_0}^{x_2} p(x) dx &= \int_{-\Delta x}^{\Delta x} A_2 x^2 + A_1 x + A_0 dx \\
&= \int_{-\Delta x}^{\Delta x} \underbrace{A_2 x^2 + A_0 dx}_{\text{even function}} + \int_{-\Delta x}^{\Delta x} \underbrace{A_1 x dx}_{\text{odd function}} \\
&= 2 \int_0^{\Delta x} A_2 x^2 + A_0 dx \\
&= \frac{2}{3} A_2 x^3 + 2 A_0 x \Big|_{x=0}^{\Delta x} \\
&= \frac{\Delta x}{3} (2 A_2 (\Delta x)^2 + 6 A_0)
\end{aligned}$$

Now

$$f(x_0) = p(-\Delta x) = A_2 (\Delta x)^2 - A_1 \Delta x + A_0$$

$$4f(x_1) = 4p(0) = 4A_0$$

$$f(x_2) = p(\Delta x) = A_2 (\Delta x)^2 + A_1 \Delta x + A_0$$

$$\Rightarrow f(x_0) + 4f(x_1) + f(x_2) = 2A_2 (\Delta x)^2 + 6A_0$$

and so

$$\begin{aligned}
\frac{1}{3} (f(x_0) + 4f(x_1) + f(x_2)) \Delta x &= \int_{x_0}^{x_2} p(x) dx \\
&\approx \int_{x_0}^{x_2} f(x) dx.
\end{aligned}$$

We have evaluated the integral of p by taking

a weight sum of $f(x_0)$, $f(x_1)$, and $f(x_2)$.

Since translations preserve area this also holds on x_0, x_1, x_2 , uniformly spaced points in $\mathbb{R}$.

Theorem #2

Let $f^{(4)}$ be continuous on $[a, b]$ and assume S_n is a Simpson's rule approximation of $\int_a^b f$.

Then

$$E_S(n) = \int_a^b f(x) \, dx - S_n$$

satisfies

$$|E_S(n)| \leq \frac{K(b-a)^5}{180n^4} = \frac{K(b-a)}{180}(\Delta x)^4.$$

where $|f^{(4)}(x)| \leq K$ for all $x \in [a, b]$.

Example #2

Estimate

$$E_T(10) \quad \text{and} \quad E_S(6)$$

for

$$\int_0^2 e^x \, dx.$$

$$f''(x) = e^x \quad |f''(x)| \leq e^2 = K$$

on $[0, 2]$ since f'' is increasing and positive on $[0, 2]$.

If $n = 10$

$$\Delta x = \frac{2-0}{10} = \frac{1}{5}$$

and so

$$|E_T(10)| \leq \frac{e^2(2-0)}{12} \left(\frac{1}{5}\right)^2 = \frac{1}{150}e^2 \approx .05$$

If $n = 6$

$$\Delta x = \frac{2-0}{6} = \frac{1}{3}$$

and so

$$|E_S(6)| \leq \frac{e^2(2-0)}{180} \left(\frac{1}{3}\right)^4 = \frac{1}{7290}e^2 \approx .001$$

since

$$|f^{(4)}(x)| \leq e^2 = K$$

on $[0, 2]$ since $f^{(4)}$ is increasing on $[0, 2]$.

Our estimates are reasonable

$$|E_S(6)| \approx .0004 \quad |E_T(10)| \approx .02$$

Example #3

Estimate

$$E_S(4) \quad \text{and} \quad E_S(6)$$

for

$$\int_0^4 x^3 \, dx.$$

$f^{(4)}(x) = 0$ on $[0, 4]$ so we have no error for $n \geq 2$ (and even)! That is,

$$|E_S(2k)| \leq 0$$

for $k \geq 1$.

Simpson's rule is exact for degree 3 (or less) polynomials. However, the method is designed to exactly integrate quadratics!

Example #4

Find

$$|E_T(8)| \quad \text{and} \quad |E_S(8)|$$

for

$$\int_0^2 e^x \, dx.$$

n	$ E_T(n) $	$ E_S(n) $
2	5×10^{-1}	3×10^{-2}

n	$ E_T(n) $	$ E_S(n) $
2	5×10^{-1}	3×10^{-2}
4	1×10^{-1}	2×10^{-3}

n	$ E_T(n) $	$ E_S(n) $
2	5×10^{-1}	3×10^{-2}
4	1×10^{-1}	2×10^{-3}
8	3×10^{-2}	1×10^{-4}

n	$ E_T(n) $	$ E_S(n) $
2	5×10^{-1}	3×10^{-2}
4	1×10^{-1}	2×10^{-3}
8	3×10^{-2}	1×10^{-4}
16	8×10^{-3}	9×10^{-6}

n	$ E_T(n) $	$ E_S(n) $
2	5×10^{-1}	3×10^{-2}
4	1×10^{-1}	2×10^{-3}
8	3×10^{-2}	1×10^{-4}
16	8×10^{-3}	9×10^{-6}
32	2×10^{-3}	5×10^{-7}

n	$ E_T(n) $	$ E_S(n) $
2	5×10^{-1}	3×10^{-2}
4	1×10^{-1}	2×10^{-3}
8	3×10^{-2}	1×10^{-4}
16	8×10^{-3}	9×10^{-6}
32	2×10^{-3}	5×10^{-7}
64	5×10^{-4}	3×10^{-8}

n	$ E_T(n) $	$ E_S(n) $
2	5×10^{-1}	3×10^{-2}
4	1×10^{-1}	2×10^{-3}
8	3×10^{-2}	1×10^{-4}
16	8×10^{-3}	9×10^{-6}
32	2×10^{-3}	5×10^{-7}
64	5×10^{-4}	3×10^{-8}
128	1×10^{-4}	2×10^{-9}

n	$ E_T(n) $	$ E_S(n) $
2	5×10^{-1}	3×10^{-2}
4	1×10^{-1}	2×10^{-3}
8	3×10^{-2}	1×10^{-4}
16	8×10^{-3}	9×10^{-6}
32	2×10^{-3}	5×10^{-7}
64	5×10^{-4}	3×10^{-8}
128	1×10^{-4}	2×10^{-9}
256	3×10^{-5}	1×10^{-10}

n	$ E_T(n) $	$ E_S(n) $
2	5×10^{-1}	3×10^{-2}
4	1×10^{-1}	2×10^{-3}
8	3×10^{-2}	1×10^{-4}
16	8×10^{-3}	9×10^{-6}
32	2×10^{-3}	5×10^{-7}
64	5×10^{-4}	3×10^{-8}
128	1×10^{-4}	2×10^{-9}
256	3×10^{-5}	1×10^{-10}
512	8×10^{-6}	8×10^{-12}